

Qatar-4b

R-2466

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-4_211228 · created 2026-08-23T01:39:05+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459576.7344 +/- 0.0024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1839 +/- 0.0083
Transit depth (Rp/Rs)^2	3.38 +/- 0.31 [%]
Orbital Inclination (inc)	87.86 +/- 1.04
Airmass coefficient 1 (a1)	1.03 +/- 0.027
Airmass coefficient 2 (a2)	-0.0247 +/- 0.0072
Scatter in the residuals of the lightcurve fit is	2.64 %
Best Comparison Star	#3 - [561, 103]
Optimal Method	PSF photometry
Transit Duration (day)	0.0933 +/- 0.0026

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.1839 ± 0.0083 vs 0.138 (deviation 4.3σ)
Duration fit vs published	0.0933 d vs 0.0892 d (deviation 1.24σ)
Mid-time O-C	-2.4 min
Depth SNR	17.2
Residual scatter	2.64 %

Light curve

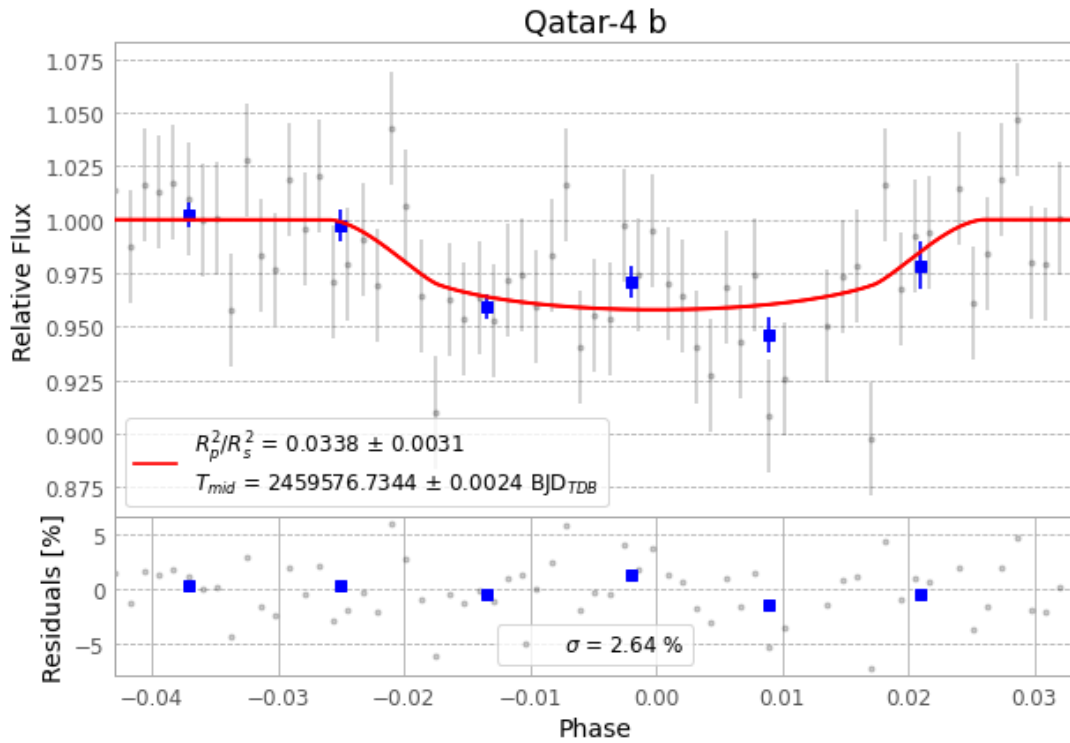


Figure 1 — FinalLightCurve_Qatar-4 b_2021-12-27.png (SHA-256 de79690d6c45c9c3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [314, 282], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0a2d934b72d6ec86...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

67 FITS frames (44 MB), Qatar-4211228034212.FITS → Qatar-4211228070012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-4-211228.log (f19b1a5b1368...), FinalLightCurve_Qatar-4 b_2021-12-27.png (de79690d6c45...), FinalLightCurve_Qatar-4 b_2021-12-27.csv (b2bee8bc2585...)

Compute resources

Wall time 97.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-23 01:39Z.